Question Paper

Physics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A diatomic ideal gas is compressed adiabatically to $\frac{1}{32}$ of its initial volume. If the initial temperature of the gas is T_i (in kelvin) and the final temperature is $T_f = aT_i$, then value of a is.....

A. 4

B. 6

C. 5

D. 9

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements are incorrect?
(i)If there were no friction, Work need to be done to move a body up an inclined plane is zero.
(ii)If there were no friction, moving vehicles could not be stopped even by locking the brakes.
(iii)As the angle of inclination is increased, the normal reaction on the body placed on it increases.
(iv)A duster weighing 0.5 kg is pressed against a vertical board with a force of 11 N. If the coefficient of friction is 0.5, the work done in rubbing it upward through a distance of 10 cm is 0.55J.

A. (i)and(ii)

B. (i),(ii),(iv)

C. (i),(iii),and(iv)

D. All of these

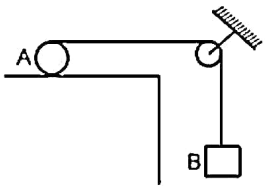
JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find the acceleration of solid right circular roller A, weighing 12 kg when it is being pulled by another weight B(6 kg) along the horizontal plane as in figure (pulley is massless). The weight B is attached to the end of a string wound around the circumference of roller. Assume there is no slipping of the roller and the string is inextensible.



A. $4\ g/7$

B. $g/7$

C. $2\ g/7$

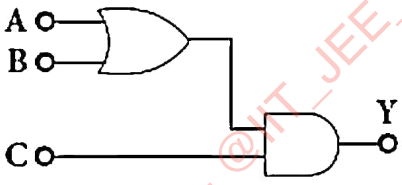
D. $3\ g/7$

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The output of the following circuit is 1 when the input ABC is



A. 101

B. 100

C. 110

D. 010

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A coil has an inductance of $5\ \mathrm{H}$ and a resistance $20\ \Omega$. An emf of $100\ \mathrm{V}$ its applied to it. What is the energy stored in the magnetic field, when the current has reached its final steady value?

A. $32.4\ \mathrm{J}$

B. $62.5\ \mathrm{J}$

C. $22.2\ \mathrm{J}$

D. $12.0\ \mathrm{J}$

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Unpolarized light of intensity $32\ \mathrm{W\ m^{-2}}$ passes through three polarizers such that transmission axes of the first and second polarizer make an angle 30° with each other and the transmission axis of the last polarizer is crossed with that of the first. The intensity of the final emerging light will be?

A. 4

B. 2.5

C. 3

D. 2

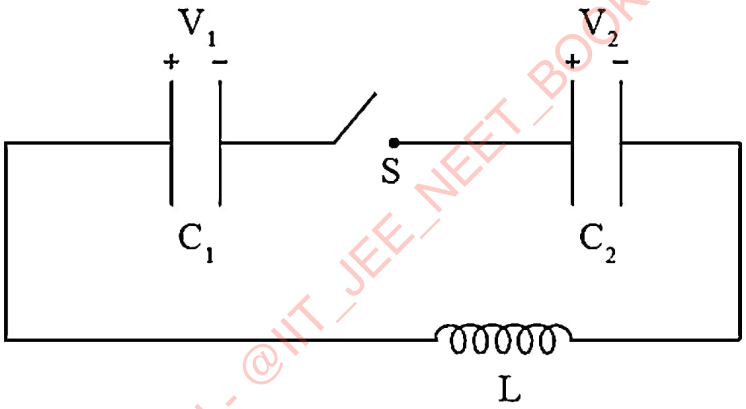
JOIN - @IIT_JEE_NEET_BOOK

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two capacitors of capacitances C_1, C_2 are charged to voltages V_1 and V_2 respectively are connected in series with the help of a switch'S' that series combination is connected to ideal inductor of inductance ' L ' as shown in figureWhen the switch is closed, then the circuit becomes L-C oscillator, find maximum possible current passing through the inductor.



A. $\sqrt{\frac{C_1V_1^2+C_2V_2^2}{L}}$

B. $|V_1 - V_2|\sqrt{\frac{C_1C_2}{(C_1+C_2)L}}$

C. $(V_1 + V_2)\sqrt{\frac{C_1C_2}{(C_1+C_2)L}}$

D. $\frac{(C_1V_1+C_2V_2)}{\sqrt{(C_1+C_2)L}}$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two planets have masses M and 16M and their radii are a and 2a, respectively. The separation between the centres of the planets is 10a. A body of mass m is fired from the surface of the larger planet towards the smaller planet along the line joining their centres. For the body to be able to reach at the surface of smaller planet, the minimum firing speed needed is :

A. $2\sqrt{\frac{GM}{a}}$

B. $4\sqrt{\frac{GM}{a}}$

C. $\sqrt{\frac{GM^2}{ma}}$

D. $\frac{3}{2}\sqrt{\frac{5GM}{a}}$

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A solenoid of 1.5 m length and 4.0 cm diameter has 10 turns per cm. A current of 5 A is flowing through it. The magnetic field induction at axis inside the solenoid is....

A. $2\pi \times 10^{-4}$ T

B. $2\pi \times 10^{-5}$ T

C. $2\pi \times 10^{-3}$ T

D. $2\pi \times 10^{-6}$ T

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The speed of a transverse wave, going on a wire having a length of 50 cm and mass 0.5 g is 80 m/s. The area of cross section of the wire is 1.0 mm^2 and its Young's modulus is 16×10^{11} N/m^2. Find the extension of the wire over its natural length (in $\times 10^{-2}$ mm).

A. 1

B. 0.5

C. 2.5

JOIN - @IIT_JEE_NEET_BOOK

D. 2

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If one mole of a monoatomic gas $\left(\gamma = \frac{5}{3}\right)$ is mixed with one mole of a diatomic gas $\left(\gamma = \frac{7}{5}\right)$ then the value of 2γ for the mixture is:

A. 4

B. 5

C. 2

D. 3

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The principal section of a glass prism is an isosceles triangle ABC with $AB = AC$. The face AC is silvered. A ray of light is incident normally on the face AB and after tworeflections, it emerges from the base BC perpendicular to the base. Angle BAC of the prism is.....

A. 60

B. 45

C. 30

D. 15

JOIN - @IIT_JEE_NEET_BOOK

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The electron in a H-atom makes transition from M shell to K shell. The ratio of magnitude of initial to final acceleration of the electron is.

A. $1 : 81$

B. $1 : 27$

C. $27 : 1$

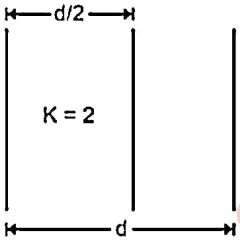
D. $1 : 9$

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A certain series $\mathrm{R} - \mathrm{C}$ circuit is formed using resistance R, a capacitor without dielectric having a capacitance $\mathrm{C} = 2\ \mathrm{F}$ and a battery of emf $E = 3\ \mathrm{V}$. The circuit is completed and it is allowed to attain the steady state. After this, at $t = 0$ half the thickness of the capacitor is filled with a dielectric constant $\mathrm{k} = 2$ as shown in the figure. The system is again allowed to attain a steady state. What will be the heat generated (in joule) in the circuit between $t = 0$ and $t = \infty$?



A. 6 J

B. 4.5 J

C. 3 J

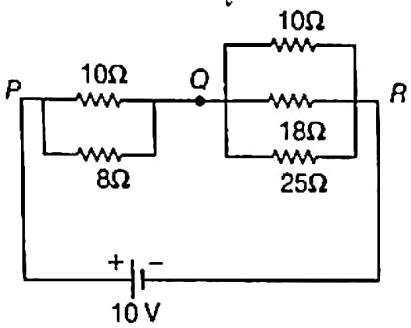
D. 1.5 J

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Find $V_P - V_Q$ in the circuit shown in figure.



A. $6.68\ V$

B. $8\ V$

C. $4.65\ V$

D. $7\ V$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the double slit experiment, 5 th dark fringe is formed opposite to one of the slits. The wavelength of light is---

A. $\frac{d^2}{6D}$

B. $\frac{d^2}{5D}$

C. $\frac{d^2}{15D}$

D. $\frac{d^2}{9D}$

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The centre of mass of three particles of masses 1 kg, 2 kg and 3 kg at $(3,\ 3,\ 3)$ with reference to a fixed coordinate system. Where should the fourth particle of mass 4 kg be placed, so that the centre of mass of the system of all particles shifts to a point $(1,\ 1,\ 1)$?

A. $(-1,\ -1,\ -1)$.

B. $(-2,\ -2,\ -2)$.

C. $(2,\ 2,\ 2)$.

D. $(1,\ 1,\ 1)$.

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

An infinite non-conducting sheet has a surface charge density of $7 \times 10^{-7} \mathrm{Cm}^{-2}$ on one side. The distance between equipotential surfaces whose potentials differ by 19.8 V, will be
(Take $\frac{1}{4\pi\epsilon_o} = 9 \times 10^9$ SI units)

A. 2.0 mm

B. 0.25 mm

C. 1.0 mm

D. 0.5 mm

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a plane EM wave the electric field oscillates sinusoidally at a frequency of 30MHz and amplitude 150 V/m. Identify the correct expression of $\overline{\mathrm{B}}$ assuming the wave is propagating along x-axis and is oscillating along y-axis.

A. $5 \times 10^{-7} \sin\left[\frac{x}{3} - 6 \times 10^{+7} t\right] \hat{z}$ T

B. $5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{5} - 6 \times 10^{+7} t\right)\right] \hat{z}$ T

C. $5 \times 10^{-7} \sin\left[\pi\left(\frac{x}{10} - 3 \times 10^{+7} t\right)\right] \hat{z}$ T

D. $5 \times 10^{-7} \sin\left[\pi\left(\frac{2x}{5} - 6 \times 10^{+8} t\right)\right] \hat{z}$ T

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A long slender rod is welded to a thin circular disc of diameter 0.5 m at a point on its circumference. The rod is in the same plane as that of the disc and forms a tangent to the disc. The radius of gyration of the disc about the rod (in m) is?

A. $\frac{1}{4}$

B. $\frac{\sqrt{5}}{8}$

C. $\frac{1}{2}$

D. $2\sqrt{2}$

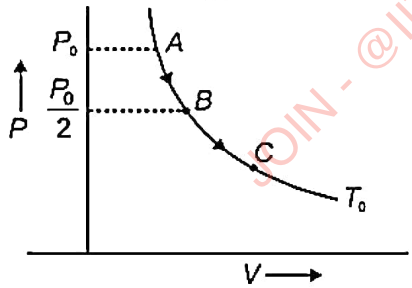
Physics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The state of an ideal gas is changed as shown, at constant temperature T_0. If heat supplied to the gas in process $B \rightarrow C$ is thrice of work done by gas in $A \rightarrow B$, then the pressure of gas in state C is $\frac{P_0}{2n}$. Find the value of n.



Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

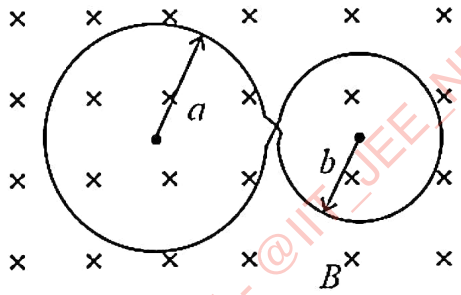
A glass flask of volume one litre at 0°C is filled, level full of mercury at this temperature. The flask and mercury are now heated to 100°C. How much mercury will spill out (in cc), if the coefficient of volume expansion of mercury is 1.82×10^{-4} C and linear expansion of glass is 0.1×10^{-4} C^{-1} respectively? (Nearest Integer)

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

A plane loop is shaped in the form, as shown in the figure with radii a $= 20$ cm $\ and\ b = 10$ cm and is placed in a uniform time-varying magnetic field
$B = B_0 \sin\omega t,$ where $B_0 = 10\ mT\ and\ \omega = 100$ rad s^{-1} .
Find the amplitude of the current induced in the loop if its resistance per unit length is equal to $50 \times 10^{-3}\ \Omega\ m^{-1}$. The inductance of the loop is negligible. (Mark answer to nearest integer)



Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

A bullet of mass 10gm is fired horizontally with velocity 1000 m/s from rifle situated at height 50 m above the ground. If the bullet reaches the ground with velocity 500 m/s, find workdone against air resistance in the trajectory of the bullet in Joules. (mark absolute value)

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

When barium is irradiated by a light of $\lambda = 4000\ \mathring{A}$, all the photo electrons emitted are bent in a circle of radius 50 cm by a magnetic field of flux density $5.26 \text{ x } 10^{-6}$ T acting perpendicular to plane of emission of photo electrons. Then, The kinetic energy of the fastest photoelectron is K eV. Find integral value of 10K.

Chemistry Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given are standard electrode potentials :
$Fe^{2+} + 2e^- \longrightarrow Fe\ ;\ E^\circ = -0.440\ V$
$Fe^{3+} + 3e^- \longrightarrow Fe\ ;\ E^\circ = -0.036\ V$
The standard electrode potential (E°) for $Fe^{3+} + e^- \longrightarrow Fe^{2+}$ is--

A. $+0.772\ V.$

B. $-0.772\ V.$

C. $+0.417\ V.$

D. $-0.414\ V.$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Two base balls (masses: $m_1 = 100$ g, and $m_2 = 50$ g) are thrown. Both of them move with uniform velocity, but the velocity of m_2 is 1.5 times that of m_1. The ratio of de Broglie wavelengths $\lambda(m_1) : \lambda(m_2)$ is given by---

A. $4 : 3$

B. $3 : 4$

C. $2 : 1$

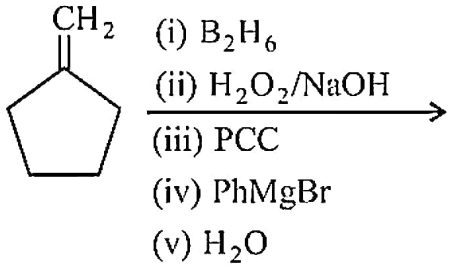
D. $1 : 2$

Question No. 3

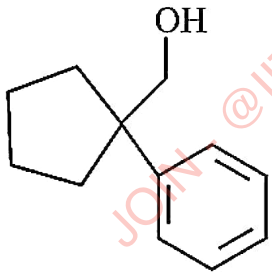
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

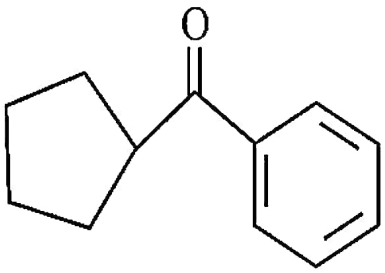
The major product of the following reactions is



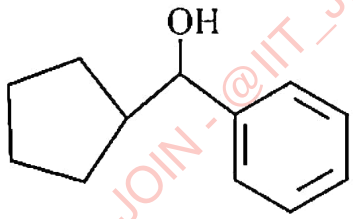
A.



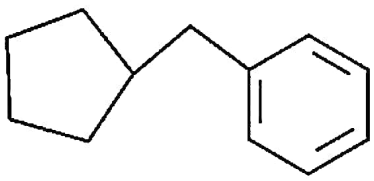
B.



C.



D.

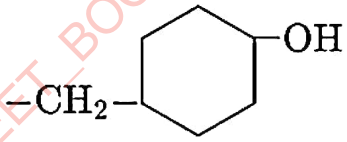


Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Match the amino acid with its correct R group.

	Amino acid		R- group
P	Cysteine	I	$-CH_2OH$
Q	Serine	II	$-CH_2-CH_2-CH_2-COOH$
R	Glutamic acid	III	
S	Tyrosine	IV	$-CH_2-SH$

A. P – I, Q – IV, R – II, S – III

B. P – IV, Q – I, R – II, S – III

C. P – III, Q – IV, R – II, S – I

D. P – II, Q – I, R – IV, S – III

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Heptane and octane form an ideal solution. At 373 K, the vapour pressures of the two liquid components are 105. 2 kPa and 46. 8 kPa, respectively. What will be the vapour pressure of a mixture of 26. 0 g of heptane and 35 g of octane?

A. 7. 308 kPa

B. 73. 08 kPa

C. 730. 8 kPa

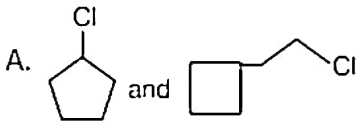
D. 7308 kPa

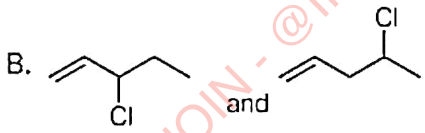
Question No. 6

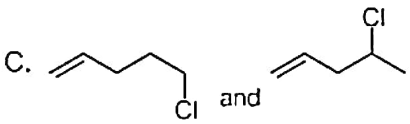
Only One Option Correct Type

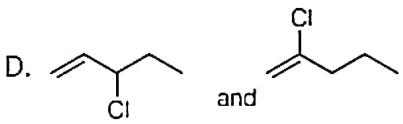
Each question has multiple options out of which ONLY ONE is correct.

X and Y are optically active isomers having formula C_5H_9Cl. When treated with one mole of H_2, X gets converted to an optically inactive compound Z, but Y gives an optically active compound P. The structure of X and Y respectively are

A. 

B. 

C. 

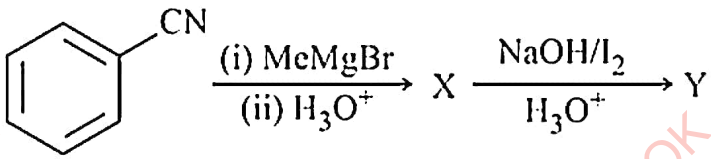
D. 

Question No. 7

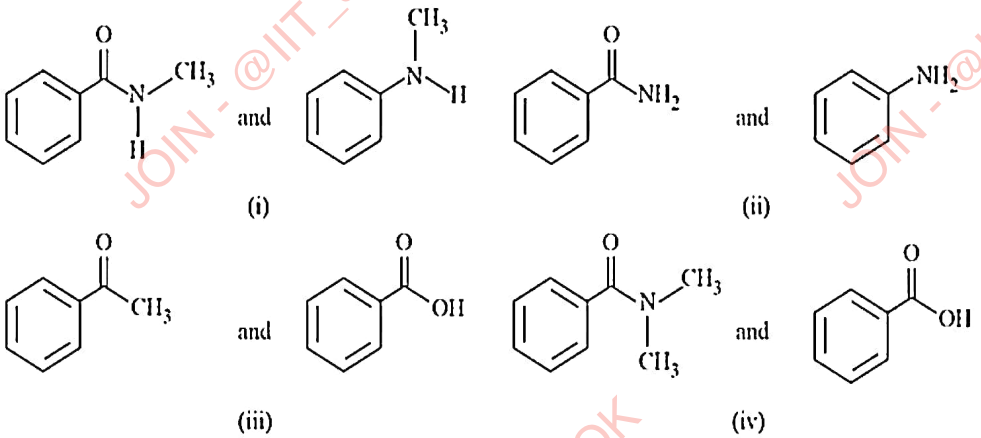
Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In the following conversion,



the major products X and Y, respectively, are



A. i

B. ii

C. iii

D. iv

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following statements was not proposed by Dalton's atomic theory?

A. Matter consists of indivisible atoms.

B. All atoms of a given element have identical properties, including identical masses.

C. Compounds can be formed when atoms of different elements combine in fractional number ratios.

D. In a chemical reaction, atoms are neither created nor destroyed.

JOIN - @IIT_JEE_NEET_BOOK

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which one of the following has highest dipole moment?

A. cis-but-2-ene

B. trans-1, 2-dichloroethene

C. cis-1, 2-dichloroethene

D. trans-but-2-ene

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The IUPAC name of the following complex is $\left[Cr(NH_3)_3(H_2O)_2Cl\right]Cl_2$.

A. Triamminediaqua chlorido chromium (III) ch!oride.

B. Diaquatriammine chlorido chromium (III) chloride.

C. Chlorido diaquatriammine chromium (III) chloride.

D. Triammine diaqua trichlorido chromium (III).

JOIN - @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Given:
(a) $n = 5, m_l = 0$
(b) $n = 4, m_l = +2, m_s = -1/2$
The maximum number of electron(s) in an atom that can have the quantum numbers as given in (a) and (b) are respectively:

A. 8 and 1

B. 10 and 2

C. 10 and 4

D. 25 and 1

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Which of the following represents the correct order of increasing first electron gain enthalpy (modulus value) for O, S, F and Cl ?

A. $\mathrm{O} < \mathrm{S} < \mathrm{F} < \mathrm{Cl}$

B. $\mathrm{S} < \mathrm{O} < \mathrm{F} < \mathrm{Cl}$

C. $\mathrm{O} < \mathrm{S} < \mathrm{Cl} < \mathrm{F}$

D. $F < S < O < Cl$

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Lucas test is used to make distinction between 1°, 2° and 3° alcohols.

$$\underset{}{ROH} + \underset{conc}{HCl} \xrightarrow{anhyd\ ZnCl_2} \underset{\substack{white\\ turbidity}}{RCl\downarrow} + H_2O$$

Which of the following statements is INCORRECT?

A. ROH behaves as base

B. Greater the value of pK_a (alcohol), greater the reactivity with conc. HCl and thus sooner the formation of white turbidity

C. Alcohol which reacts fastest with Na metal, will give turbidity at fastest rate.

D. Alcohol which gives red colour during Victor Meyer test will give turbidity at slower rate than those giving blue or white colour during Victor Meyer test.

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The increasing order of acidic strength among the following compounds is

I. Benzoic acid
II. 4 —nitrobenzoic acid
III. 3, 4 —dinitrobenzoic acid
IV. 4 —methoxybenzoic acid

A. $I < II < III < IV$

B. $I < IV < II < III$

C. $IV < I < II < III$

D. $IV < I < III < II$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

1 mole of N_2H_4 loses 10 moles of electrons to form a new compound Y. Assuming that all the nitrogen appears in the new compound, what is the oxidation state of nitrogen in Y? (There is no change in the oxidation state of hydrogen).

A. -1

B. -3

C. $+3$

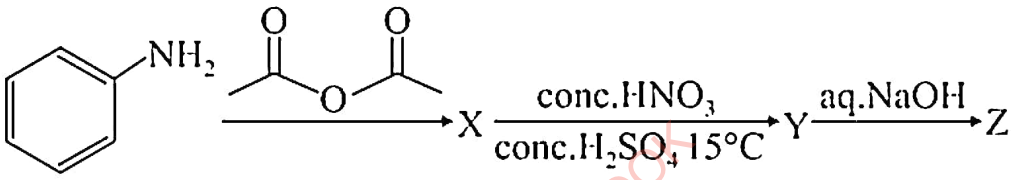
D. $+5$

Question No. 16

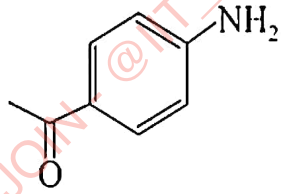
Only One Option Correct Type

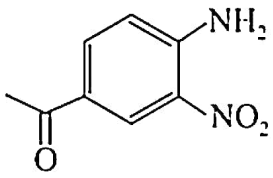
Each question has multiple options out of which ONLY ONE is correct.

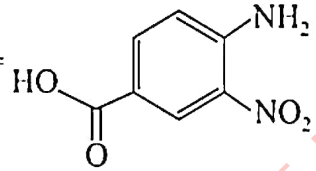
The major products X, Y and Z in the following sequence of transformations

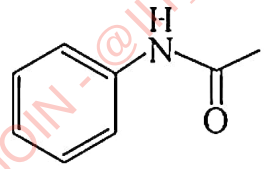


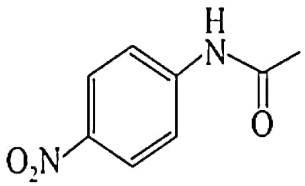
are:

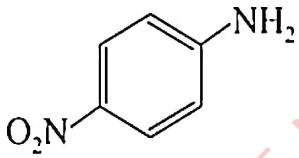
X = 

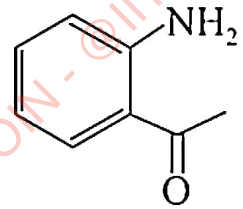
Y = 

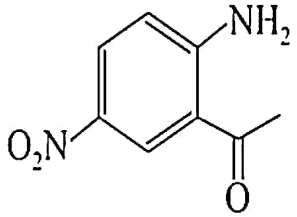
A. Z = 

X = 

Y = 

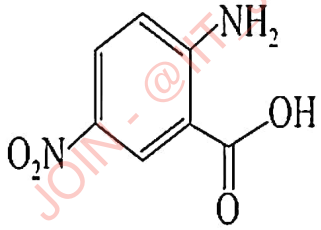
B. Z = 

X = 

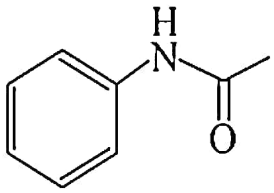
Y = 

JOIN - @IIT_JEE_NEET_BOOK

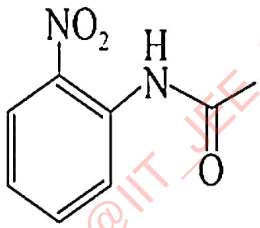
C. Z =



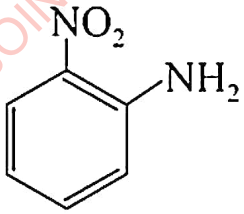
X =



Y =



D. Z =



Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The correct order of ionic radii of Ce, La, Pm and Yb in +3 oxidation state is

A. $Yb^{3+} < Pm^{3+} < Ce^{3+} < La^{3+}$

B. $La^{3+} < Ce^{3+} < Pm^{3+} < Yb^{3+}$

C. $Yb^{3+} < Ce^{3+} < Pm^{3+} < La^{3+}$

D. None of these

JOIN - @IIT_JEE_NEET_BOOK

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The carbonyl metal complex which cannot exist:

A. $Cr(CO)_6$

B. $Mn(CO)_5$

C. $Ni(CO)_4$

D. $Fe(CO)_5$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Assertion (A) : 1- Bromopentane reacts with $AgCN$ to give pentylisocyanide

Reason (R): $AgCN$ is mainly ionic in nature

A. A is true, R is true and R is correct explanation of A

B. A is true, R is true but R is not correct explanation of A

C. A is true, R is false

D. A is false, R is true

JOIN - @IIT_JEE_NEET_BOOK

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The interatomic forces of attraction present in noble gas are _____.

A. dipole-dipole interaction

B. ion-dipole interaction

C. induced dipole-induced dipole interaction

D. ion-induced dipole interaction

Chemistry Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

Among the following, the number of compounds having all atoms in a single plane are

$\mathbf{XeF_4}, \mathbf{SF_4}, \mathbf{Al_2Cl_6}, \mathrm{SnCl_4}, \mathbf{I_2Cl_6}, \mathbf{H_2C=C=C=CH_2}, \mathrm{PhCl}, \mathbf{C_2(CN)_4}, \mathrm{ICl_3}, \mathrm{PCl_5}$

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

From the following number of fat soluble and water soluble vitamins respectively are X and Y respectively, List of vitamins: Vitamin A, Vitamin D, Vitamin C, Vitamin B, Vitamin K. Find the value of $2X + 5Y$.

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

For a reaction $A + B \longrightarrow P$, the following data are provided

-

Entry	[A] in M	[B] in M	Initial rate (M/s)
1	0.02	0.02	2×10^{-2}
2	0.02	0.04	4×10^{-2}
3	0.04	0.04	8×10^{-2}

The rate constant for this reaction in standard unit is

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Using the data provided, calculate the multiple bond energy $\left(\mathrm{kJmol}^{-1}\right)$ of $\mathrm{C} \equiv \mathrm{C}$ bond in $\mathrm{C_2H_2}$. That energy is (take the bond energy of a $\mathrm{C-H}$ bond as $350\ \mathrm{kJ\ mol^{-1}}$)

$2\mathrm{C(s)} + \mathrm{H_2(\ g)} \longrightarrow \mathrm{C_2H_2(\ g)}; \Delta\mathrm{H} = 225\ \mathrm{kJ\ mol^{-1}}$

$2\mathrm{C(s)} \longrightarrow 2\mathrm{C(g)}; \Delta\mathrm{H_{sub}} = 1410\ \mathrm{kJ}$

$\mathrm{H_2(\ g)} \longrightarrow 2\mathrm{H(g)}; \Delta\mathrm{H} = 330\ \mathrm{kJ\ mol^{-1}}$

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

$20\mathrm{ml}$ of $0.1\ \mathrm{M}\ \mathrm{H_2SO_4}$ solution is added to $30\mathrm{ml}$ of $0.2\ \mathrm{M}\ \mathrm{NH_4OH}$ solution. The pH of the resultant mix. is : $(\mathrm{pK_b}$ of $\mathrm{NH_4OH} = 4.7)(\log 2 = 0.3)$

Mathematics Single Correct (Maximum Marks: 80)

Question No. 1

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f(x) = \log_e \left(e^{2x} \left(\frac{3x+5}{5-3x} \right)^{\frac{2}{3}} \right)$, $x \neq \frac{-5}{3}, \frac{5}{3}$, then the value of $\frac{df}{dx}$ at $x = 1$, is

A. $\frac{5}{4}$

B. $\frac{7}{4}$

C. $\frac{11}{4}$

D. $\frac{13}{4}$

Question No. 2

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The sum of four numbers in A.P. is 48, and the product of the extremes is to the product of the two middle terms as 27 to 35. The largest term of the A. P. is

A. 10

B. 12

C. 14

D. 18

JOIN - @IIT_JEE_NEET_BOOK

Question No. 3

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $f : R \to R$ is defined as $f(x) = \begin{cases} x+4, & x < -4 \\ 3x+2, & -4 \le x < 4, \\ x-4, & x \ge 4 \end{cases}$ then the value of $f(f(f(f(0)))) + 1$ is equal to:

A. 0

B. 1

C. 2

D. 4

Question No. 4

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $\mathrm{F(x)}$ be a non-negative continuous function defined on R such that $\mathrm{F(x)} + \mathrm{F}\left(\mathrm{x} + \frac{1}{2}\right) = 1$. Then value of $\frac{1}{1000}\int_0^{1500} F(x)dx$

A. 0.5

B. 1.5

C. 0.75

D. 0.25

JOIN - @IIT_JEE_NEET_BOOK

Question No. 5

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of five-digit numbers formed with the digits $0, 1, 2, 3, 4$ and 5 (without repetition) and divisible by 6 are

A. 72

B. 84

C. 96

D. 108

Question No. 6

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

In a ΔABC, AD bisects angle A and D lies on BC. Length of AD when $\overrightarrow{AB} = 2i + 2j + k$ and $\overrightarrow{AC} = 3i + 4k$ is:

A. $\frac{3}{8}\sqrt{30}$

B. $\frac{5}{8}\sqrt{30}$

C. $\frac{1}{8}\sqrt{30}$

D. $\frac{\sqrt{30}}{4}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 7

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\lim_{x \to 0} \left(x^{-3} \sin 3x + ax^{-2} + b \right)$ exists and is equal to 0, then

A. $a = -3$ and $b = 9/2$

B. $a = 3$ and $b = 9/2$

C. $a = -3$ and $b = -9/2$

D. $a = 3$ and $b = -9/2$

Question No. 8

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x) = x^2 + ax + b$. If $\forall x \in R$, there exist a real value of y such that $f(y) = f(x) + y$, then find the maximum value of $100a$

A. 50

B. 200

C. 120

D. 100

JOIN - @IIT_JEE_NEET_BOOK

Question No. 9

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The area of the region $\{(x,\ y) : x^2 + y^2 \leq 1 \leq x + y\}$ is

A. $\frac{\pi^2}{5}$ sq unit

B. $\frac{\pi^2}{2}$ sq unit

C. $\left(\frac{\pi}{4} - \frac{1}{2}\right)$ sq unit

D. $\frac{\pi}{4}$ sq unit

Question No. 10

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos t & \sin t \\ 0 & -\sin t & \cos t \end{pmatrix}$

Let $\lambda_1, \lambda_2, \lambda_3$ be the roots of $\det(A - \lambda I_3) = 0$, where I_3 denotes the identity matrix. If $\lambda_1 + \lambda_2 + \lambda_3 = \sqrt{2} + 1$, then the set of possible values of t, $-\pi \leq t < \pi$ is

A. a void set

B. $\left\{\frac{\pi}{4}\right\}$

C. $\left\{-\frac{\pi}{4}, \frac{\pi}{4}\right\}$

D. $\left\{-\frac{\pi}{3}, \frac{\pi}{3}\right\}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 11

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $f(x)$ be a function with continuous derivative on the interval (λ, μ) and $2\ln\left(f(x)+\sqrt{(f(x))^2+1}\right)\cdot f'(x)-\sqrt{f^2(x)+1}\leq 0$ then find minimum value of $(\mu-\lambda)$ if $\text{Lim}_{x\to\lambda^+} f(x)=-3$ and $\text{Lim}_{x\to\mu^-} f(x)=3$

A. 0

B. 1

C. -1

D. None of these

Question No. 12

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If $\int \frac{x^2}{x^4+3x^2+9}\,dx = A\tan^{-1}\left(\frac{x^2-3}{3x}\right)+\frac{B}{\sqrt{3}}\ln\left|\frac{x^2-\sqrt{3}x+3}{x^2+\sqrt{3}x+3}\right|+c$, then find the value of $12(A+B)$

A. 7

B. 10

C. 5

D. 14

Question No. 13

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The eccentricity of the ellipse which meets the straight line $\frac{x}{7}+\frac{y}{2}=1$ on the axis of x and the straight line $\frac{x}{3}-\frac{y}{5}=1$ on the axis of y and whose axes lie along the axes of coordinates is $\frac{2\sqrt{6}}{\lambda}$. Find λ

A. 5

B. 7

C. 4

D. 9

Question No. 14

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

A biased coin is tossed repeatedly until a tail appears for the first time. Heads is 3 times as likely to appear as tails. Let x be the number of tosses required. Assume that all the trials of tossing a biased coin are independent. Then, the conditional probability that $x \geq 6$, given that $x > 3$, is equal to

A. $\frac{9}{16}$

B. $\frac{1}{16}$

C. $\frac{1}{2}$

D. $\frac{3}{5}$

Question No. 15

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let the line $y = \log(a)(a > 0, a \neq 1)$ intersects the line $y = x$ at point P. If point P lies on parabola $y^2 = x$, then focal distance of point P is -

A. e

B. $e + \frac{1}{4}$

C. 1

D. $\frac{5}{4}$

Question No. 16

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

If a, x are real numbers and $|a| < 1, |x| < 1$ then $1(1+a)x + \left(1+a+a^2\right)x^2 + \ldots\infty$ is equal to

A. $\frac{1}{(1-a)(1-ax)}$

B. $\frac{1}{(1-a)(1-x)}$

C. $\frac{1}{(1-x)(1-ax)}$

D. $\frac{1}{(1+ax)(1-a)}$

JOIN - @IIT_JEE_NEET_BOOK

Question No. 17

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

The number of solution(s) of the equation $z^2 = 4z + |z|^2 + \frac{16}{|z|^3}$ is (where $z = x + iy,\ x, y \in R, i^2 = -1$ and $x \neq 2$)

A. 0

B. 1

C. 2

D. 3

Question No. 18

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Let $A = [a_{ij}]_{3\times 3}$ be a square matrix such that $AA^T = 4I, |A| < 0$. If $\begin{vmatrix} a_{11}+4 & a_{12} & a_{13} \\ a_{21} & a_{22}+4 & a_{23} \\ a_{31} & a_{32} & a_{33}+4 \end{vmatrix} = 5\lambda|A+I|$. Then, λ is equal to

A. $\frac{4}{5}$

B. $-\frac{4}{5}$

C. $\frac{8}{5}$

D. $-\frac{8}{5}$

Question No. 19

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider a function $f(\mathrm{x})$ which satisfies $f'(\mathrm{x}) - \tan \mathrm{x} \cdot f(\mathrm{x}) = \int_{-\pi/6}^{\pi/6} f(\mathrm{x})\mathrm{dx} \forall \mathrm{x} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ and $f(0) = 1$. The value of $\left(f\left(\frac{\pi}{3}\right) - \sqrt{3}\ln 3\right)$ is

A. 8

B. 4

C. 2

D. 6

Question No. 20

Only One Option Correct Type

Each question has multiple options out of which ONLY ONE is correct.

Consider, $f(x) = \left(\lambda^4 - 2\lambda^2\right)x + 3\cos x, \lambda \in R.$
The least positive integral value of λ for which $f(x)$ is monotonically increasing for all $x \in R.$

A. 2

B. 3

C. 4

D. 5

Mathematics Numerical (Maximum Marks: 20)

Question No. 1

Numerical Type

The answer has to be filled into the input box provided below.

The number of points at which the function $f(x) = (x - |x|)^2(1 - x + |x|)^2$ is not differentiable in the interval $(-3, 4)$ is

Question No. 2

Numerical Type

The answer has to be filled into the input box provided below.

Suppose $(5x - 4y)^{23} = \sum_{r=0}^{23} T_{r+1}$, where $T_{r+1} = {}^{23}C_r\ (5x)^{23-r}\ (-4y)^r$ and let $x = \frac{1}{6}$ and $y = \frac{1}{8}$. Then, what will be the possible value of $\sum_{r=1}^{23} \text{sgn}(|T_r| - |T_{r+1}|)$

Question No. 3

Numerical Type

The answer has to be filled into the input box provided below.

In $\triangle ABC$, a point P is chosen on side $\overrightarrow{AB}$ so that $AP : PB = 1 : 4$ and a point Q is chosen on the side $\overrightarrow{BC}$ so that $CQ : QB = 1 : 3$. Segment $\overrightarrow{CP}$ and $\overrightarrow{AQ}$ intersect at M. If the ratio $\frac{MC}{PC}$ is expressed as a rational number in the lowest term as $\frac{a}{b}$, then find $(a + b)$.

Question No. 4

Numerical Type

The answer has to be filled into the input box provided below.

Let $x_1, x_2, \ldots\ldots\ldots, x_{100}$ are 100 observations such that $\sum x_i = 0, \sum_{1 \le i < j \le 100} |x_i x_j| = 80000$ & mean deviation from their mean is 5, then their standard deviation, is-

Question No. 5

Numerical Type

The answer has to be filled into the input box provided below.

The straight lines $7x - 2y + 10 = 0$ and $7x + 2y - 10 = 0$ form an isosceles triangle with the line $y = 2$. Area of this triangle is equal to $\frac{2\lambda}{\mu}$ sq. units. Then, the value of $\lambda - \mu$ is

JOIN - @IIT_JEE_NEET_BOOK